

Class 10 • Practice Test (Maths and Science)

Time: 1 hour • Maximum marks: 25 • Answer all questions. Show your working.

Q1.	The HCF of 6 and 20 is: (a) 1 (b) 2 (c) 4 (d) 120	[1]
Q2.	Find the roots of the quadratic equation $2x^2 - 7x + 3 = 0$ by factorisation.	[2]
Q3.	Find the sum of the first 20 terms of the AP: 3, 7, 11, 15, ...	[3]
Q4.	If $\sin A = 3/5$, find the values of $\cos A$, $\tan A$ and $\sec A$.	[3]
Q5.	Three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in parallel to a 6 V battery. Find (i) the equivalent resistance of the combination and (ii) the current drawn from the battery.	[3]
Q6.	Find the coordinates of the point which divides the line segment joining $(2, -3)$ and $(5, 6)$ internally in the ratio 2 : 1.	[2]
Q7.	Balance the following chemical equation and state the type of reaction: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$	[2]
Q8.	Why is the wall of the left ventricle of the human heart thicker than that of the right ventricle?	[2]
Q9.	(i) Two dice are thrown together. Find the probability of getting a sum of 7. [2] (ii) A bag contains 5 red and 3 blue balls. One ball is drawn at random. Find the probability that it is blue. [1]	[3]
Q10.	An object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the position of the image.	[3]